



Instituto Infnet

Disciplina - Desenvolvimento FrontEnd Com Streamlit

Aplicação de Soccer Analytics

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Importação de bibliotecas

```
import streamlit as st
from statsbombpy import sb
import pandas as pd
import matplotlib.pyplot as plt
```



statsbomppy

- Para download dos dados

Seleção de competição

```
df_comp = sb.competitions()
df_comp_nat =
df_comp[df_comp['competition_international'] ==
False]
df_comp_nat[['year1', 'year2']] =
df_comp_nat['season_name'].str.split('/',
expand=True)
competition_names =
df_comp_nat['competition_name'].unique()

competition_selected = st.selectbox("Competição: ",
competition_names)
```



'competition_international'] == False

- Apenas competições nacionais

Seleção da temporada

```
df_comp_sel =  
df_comp_nat[df_comp_nat['competition_name'] ==  
competition_selected]  
  
season_names =  
df_comp_sel.sort_values('year1')['season_name']  
season_selected = st.selectbox("Temporada: ",  
season_names)
```



```
.sort_values('year1')['season_name']
```

- Mostrar os anos em ordem

Seleção das partidas e times

```
df_comp_season_sel =  
df_comp_sel[(df_comp_sel['competition_name'] ==  
competition_selected) & \  
  
(df_comp_sel['season_name'] == season_selected)]  
  
matches =  
sb.matches(season_id=df_comp_season_sel['season_id']  
  
competition_id=df_comp_season_sel['competition_id']  
.item())  
  
teams = matches['home_team'].unique()  
team_selecteds = st.multiselect ("Times: ", teams)  
  
if len(team_selecteds) <= 0:  
    st.stop()
```

Gráfico Evolução de Gols Marcados

```
st.subheader("Evolução de Gols")
fig, ax = plt.subplots()
for i, team_selected in enumerate(team_selecteds):
    matches_team_sel = matches[(matches['home_team'] == team_selected) |
    (matches['away_team'] == team_selected)]
    matches_team_sel['match_date'] =
pd.to_datetime(matches_team_sel['match_date'])
    matches_team_sel = matches_team_sel.sort_values('match_date')
    matches_team_sel = matches_team_sel.reset_index()

    matches_team_sel['all_score'] = matches_team_sel.apply(lambda row:
row['home_score'] if row['home_team'] == team_selected else row['away_score'],
axis=1)
    matches_team_sel['all_score_cumsum'] =
matches_team_sel['all_score'].cumsum()

    ax.plot(matches_team_sel.index, matches_team_sel['all_score_cumsum'],
marker='o', label=team_selected)

ax.legend()
st.pyplot(fig)
```

Gráfico Distribuição de Gols Marcados

```
st.subheader("Distribuição de gols")
cols = st.columns(len(team_selecteds))
for i, team_selected in enumerate(team_selecteds):
    fig2, ax2 = plt.subplots()
    ax2.hist(matches_team_sel['all_score'])
    ax2.set_title(team_selected)
    ax2.set_xlabel('Total de gols')
    ax2.set_ylabel('Frequência')

    cols[i].pyplot(fig2)
```